

Literature Review

Multi-View Mammographic Breast Cancer Detection with Bilateral Asymmetry Fusion and Conformal Uncertainty Quantification

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Abstract

This literature review surveys the research foundations underpinning a novel deep learning system for breast cancer detection from digital mammography. The proposed system makes two primary methodological contributions: multi-view bilateral asymmetry fusion, implemented via a Siamese encoder with gated cross-view attention and an explicit asymmetry module $\delta = |f_L - f_R|$; and conformal uncertainty quantification, providing distribution-free finite-sample coverage guarantees via Regularised Adaptive Prediction Sets (RAPS). The review covers ten primary sources organised across four thematic areas: the backbone architecture and fine-tuning paradigm that underpin the feature extractor; the mammography AI literature that establishes clinical credibility and defines the experimental baselines; the multi-view and bilateral asymmetry methods that constitute the immediate prior work for the first contribution; and the conformal prediction literature that grounds the second. Each paper is assessed for its direct methodological relevance to the proposed system, with explicit attention to what the proposed work extends, what it must differentiate itself from, and what it implements directly.

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Introduction

Breast cancer is the most commonly diagnosed cancer among women worldwide. Mammographic screening programmes reduce mortality through early detection, yet sensitivity is constrained by reader variability, radiologist fatigue, and the difficulty of identifying subtle lesions in radiographically dense tissue. Deep convolutional neural networks have demonstrated radiologist-level diagnostic performance on large mammography datasets (McKinney et al., 2020; Wu et al., 2019), catalysing substantial clinical and research interest in automated screening.

The proposed system targets two underaddressed limitations in the current literature. First, most high-performing systems process views independently or fuse them implicitly, discarding the well-established clinical value of bilateral asymmetry — the systematic structural difference between left and right breasts that is one of the most reliable radiological indicators of malignancy. Second, predictive uncertainty in deployed mammography AI is routinely characterised by heuristic methods lacking formal guarantees, a critical shortcoming in a clinical decision-support context where the cost of overconfident errors is asymmetric and severe.

The proposed system addresses these limitations through a Siamese bilateral encoder with gated cross-view attention, an explicit asymmetry feature $\delta = |f_L - f_R|$, and split conformal prediction via RAPS (Romano et al., 2020), providing marginal coverage guarantees at any user-specified miscoverage level α . This review surveys the ten primary sources that establish the intellectual context, methodological precedents, and comparative baselines for the system. The review is structured in four thematic sections, each corresponding to a load-bearing component of the proposed architecture. All citations follow APA 7th Edition, as standard for quantitative and computational research submissions at University College Dublin.

Backbone Architecture and the Fine-Tuning Paradigm

DenseNet: The Architectural Foundation

The proposed system employs DenseNet-121 as its primary feature extraction backbone. Huang et al. (2017) introduced dense connectivity, in which each layer receives the concatenated fea-

ture maps of *all* preceding layers as input, rather than only the immediately preceding layer as in residual networks. This connectivity pattern promotes feature reuse, substantially reduces parameter count relative to comparably deep networks, and produces implicit deep supervision throughout the network by shortening the gradient path from any layer to the loss. DenseNet-121 — the 121-layer configuration — achieves strong performance on ImageNet at a parameter budget well-suited to fine-tuning on datasets of limited size, making it the natural choice for CBIS-DDSM (Lee et al., 2017), which contains approximately 2,620 cases. Understanding the dense connectivity structure is a prerequisite for reasoning about feature map dimensions and skip connections in the bilateral fusion module, where the concatenation of encoder outputs from both branches is combined with the asymmetry feature δ .

CheXNet and the Medical Fine-Tuning Template

Rajpurkar et al. (2017) fine-tuned DenseNet-121 on the ChestX-ray14 dataset of 112,120 frontal chest radiographs, achieving F1 scores exceeding the mean radiologist performance on pneumonia detection. Their protocol — ImageNet-pretrained initialisation, aggressive data augmentation (random horizontal flipping, random crop), and Adam optimisation with a cyclically decayed learning rate — established the canonical transfer learning template for medical imaging with DenseNet. This protocol is adopted directly in the proposed system’s training pipeline. CheXNet also introduced the practice of reporting performance at the per-finding level with radiologist comparison, a reporting standard that the proposed system’s evaluation follows for its CBIS-DDSM results.

Mammography AI: Clinical Baselines

DeepMind International Evaluation

McKinney et al. (2020) evaluated a proprietary Google DeepMind mammography system on 29,426 retrospective cases from the United Kingdom and United States, supplemented by a prospective UK cohort. The system reduced false positive rates by 5.7% (UK) and 1.2% (US) and false negative rates by 9.4% (UK) and 2.7% (US) relative to a single radiologist, outperforming all six radiologists who participated in an independent reader study. This paper establishes

the performance expectations and clinical framing for the field. It is the reference every examining supervisor will situate the proposed work against, and its results define the upper bound of clinical credibility against which new systems are benchmarked. The proposed system does not attempt to replicate the scale of this study — it trains and evaluates on publicly available data (CBIS-DDSM) — but positions its contributions relative to the architectural and methodological directions this work opened.

NYU Deep Learning for Breast Cancer Screening

Wu et al. (2019) presented the most methodologically detailed academic mammography study to date, training on 229,426 screening examinations from NYU Langone Health. Their model achieved AUROC 0.876 on an independent test set and produced statistically significant improvements in radiologist sensitivity when used as a second reader. Critically for the proposed system, Wu et al. (2019) process all four standard views — left and right craniocaudal (CC) and mediolateral oblique (MLO) — and combine view-level predictions through learned attention weights. This is the direct architectural template for the proposed system. The NYU open-source codebase provides the reference implementation for the single-view DenseNet-121 baseline against which the bilateral asymmetry fusion component is evaluated. The proposed system extends Wu et al. by replacing implicit view combination with a Siamese bilateral encoder, a gated cross-view attention module, and an explicit asymmetry feature, and by adding conformal prediction sets at inference time.

CBIS-DDSM: The Primary Dataset

Lee et al. (2017) describe the Curated Breast Imaging Subset of the Digital Database for Screening Mammography (CBIS-DDSM), the primary dataset on which the proposed system is trained and evaluated. CBIS-DDSM contains 2,620 digitised film-screen mammography cases with expert annotations for mass and calcification findings, BI-RADS assessment categories, pathology labels (benign/malignant), bounding boxes, and pixel-level segmentation masks for a subset of cases. Reading this paper is a prerequisite before writing any dataloader code. The dataset's structure has several non-obvious properties that affect implementation: cases are stored as

DICOM files in a non-standard directory layout; image-level and case-level CSV files must be joined on `patient_id` and laterality to reconstruct bilateral pairs; and the train/test split is provided at the case level rather than the image level, requiring careful handling to avoid data leakage between left-right pairs. The proposed system’s bilateral fusion module requires precisely this patient-level grouping, making the data structure described by Lee et al. (2017) architecturally relevant, not merely bibliographically obligatory.

Multi-View and Bilateral Asymmetry Fusion

Dual-View Deep Learning for Mammography

Wu et al. (2020) extended the single-view NYU architecture into an explicit dual-view framework, processing CC and MLO views of the same breast jointly through a shared encoder with a cross-view attention mechanism. Their system achieved statistically significant improvements over the single-view baseline on the NYU proprietary dataset, demonstrating that explicit architectural modelling of the CC/MLO relationship carries diagnostic information beyond what is captured by independent view processing with late fusion. This paper is the most direct architectural prior work for the proposed system and must be read with particular care. The key dimension along which the proposed system advances beyond Wu et al. (2020) is the *bilateral* axis: their model fuses views of the same breast (CC and MLO), whereas the proposed system additionally fuses across the left and right breast, treating the contralateral pair as a source of discriminative asymmetry signal. This distinction is the primary claim of novelty and must be stated precisely in every comparison with this paper.

Multi-View Bilateral Asymmetry Detection

Nguyen et al. (2019) proposed the most direct computational precursor to the proposed system’s bilateral asymmetry module. They trained a shared-weight CNN encoder on bilateral mammogram pairs and computed an explicit asymmetry feature as the absolute difference between left and right breast representations, $\delta = |f_L - f_R|$, which was then passed to a classifier jointly with the individual view features. Their architecture demonstrated that this asymmetry signal improved classification AUROC by a statistically significant margin over a unilateral

baseline on CBIS-DDSM. The proposed system directly adopts the $\delta = |f_L - f_R|$ formulation and the shared-weight Siamese encoder structure, extending it with a gated cross-view attention mechanism that modulates the contribution of each view representation based on the asymmetry magnitude, and with the conformal prediction wrapper at inference.

Bilateral Asymmetry Counterfactual Network

Yang et al. (2021) proposed a bilateral asymmetry guided counterfactual generating network, in which the asymmetry signal between left and right breast encodings is used to synthesise a plausible normal counterpart for an abnormal breast. Classification is then performed contrastively between the real and counterfactual representations. Their system achieves strong performance and demonstrates the diagnostic power of the bilateral asymmetry signal when exploited as a generative prior. This paper is included not as a precursor to extend but as a comparison point to differentiate from. The proposed system and Yang et al. (2021) share the core insight that bilateral asymmetry is a powerful discriminative signal, but differ fundamentally in mechanism: the proposed system incorporates asymmetry as a direct discriminative feature fused through attention, preserving interpretability via standard gradient-based visualisation, whereas Yang et al. (2021) use asymmetry as a generative prior, introducing substantial additional architectural complexity and a generative training objective. Being precise about this distinction is necessary to establish the novelty of the proposed contribution.

Conformal Prediction and Uncertainty Quantification

Distribution-Free Uncertainty Quantification

Angelopoulos and Bates (2021) provide the definitive tutorial introduction to conformal prediction and distribution-free uncertainty quantification, covering the theoretical foundations, split and full conformal variants, and the construction of prediction sets with finite-sample coverage guarantees. This paper grounds the proposed system’s second primary contribution. Split conformal prediction — the variant implemented — partitions available data into training, calibration, and test sets. Nonconformity scores are computed on the calibration set, and the $(1-\alpha)(1+1/n)$ -th quantile of these scores serves as the threshold for prediction set construction

at test time. The resulting sets satisfy a marginal coverage guarantee: for any user-specified miscoverage level $\alpha \in (0, 1)$, the prediction set contains the true class with probability at least $1 - \alpha$, regardless of the underlying data distribution. This distribution-free property is the defining advantage of conformal prediction over calibration approaches such as temperature scaling, which improve reliability on average but provide no per-instance statistical guarantee. In a clinical screening context — where the consequences of overconfident false negatives are severe and asymmetric — the formal guarantee is not merely a theoretical nicety but a prerequisite for principled deployment.

Regularised Adaptive Prediction Sets (RAPS)

Romano et al. (2020) introduced Regularised Adaptive Prediction Sets (RAPS), extending the basic split conformal framework with a regularisation term that penalises prediction sets containing many low-probability classes. Standard split conformal prediction constructs sets of fixed threshold, which can be arbitrarily large when the model is uncertain. RAPS addresses this by adding a size-regularisation penalty $\lambda \cdot \max(0, |\mathcal{C}| - k_{\text{reg}})$ to the nonconformity score, where $|\mathcal{C}|$ is the current set size and k_{reg} is the target size. The penalty encourages tight prediction sets while preserving the marginal coverage guarantee. RAPS is the specific conformal variant implemented in the proposed system, and its hyperparameters $(\lambda, k_{\text{reg}})$ are tuned on the CBIS-DDSM calibration split. In the clinical context, the practical consequence is that the system returns tight prediction sets for confident predictions (typically a singleton $\{\text{benign}\}$ or $\{\text{malignant}\}$) while appropriately expanding to a two-element set $\{\text{benign}, \text{malignant}\}$ for ambiguous cases, directly encoding the clinical recommendation to recall for further assessment.

Synthesis

The ten papers reviewed above collectively define the proposed system’s intellectual heritage across four dimensions.

The backbone and fine-tuning papers (Huang et al., 2017; Rajpurkar et al., 2017) establish that DenseNet-121 with ImageNet pre-training is the appropriate feature extractor for a dataset of CBIS-DDSM’s size, and that the CheXNet training protocol is the correct initialisa-

tion strategy.

The clinical baseline papers (Lee et al., 2017; McKinney et al., 2020; Wu et al., 2019) define the field’s performance expectations, provide the direct architectural template the proposed system extends, and specify the dataset whose structure determines the bilateral pairing logic.

The bilateral asymmetry papers (Nguyen et al., 2019; Wu et al., 2020; Yang et al., 2021) establish the immediate prior work. Nguyen et al. (2019) is the system directly extended; Wu et al. (2020) is the closest published architecture; Yang et al. (2021) is the paper from which the proposed system must differentiate itself most carefully.

The conformal prediction papers (Angelopoulos & Bates, 2021; Romano et al., 2020) provide the theoretical foundation and specific implementation of the second primary contribution. Together they establish both why conformal prediction is the appropriate uncertainty quantification framework in a clinical context (formal coverage guarantees, distribution-free validity) and exactly which variant the proposed system implements and why (RAPS for efficient, size-regularised prediction sets).

No existing system combines bilateral asymmetry fusion, explicit δ -feature modulation via gated cross-view attention, and distribution-free conformal uncertainty quantification within a unified architecture trained on publicly available mammography data. This combination constitutes the novel contribution the proposed work makes to the field.

Conclusion

This review has surveyed ten primary sources across four thematic areas directly relevant to the proposed system. The backbone and transfer learning literature establishes the feature extractor and training protocol. The mammography AI literature defines clinical baselines, architectural templates, and the primary dataset. The bilateral asymmetry literature provides the immediate precursors that the proposed system extends and must differentiate from. The conformal prediction literature grounds the uncertainty quantification contribution in theory and specifies the implemented method.

The scope of this review is intentionally narrow, covering only those works that are

load-bearing for the system's two primary contributions. Each paper is cited because it directly informs an architectural decision, provides a baseline the system must surpass, or must be engaged with explicitly to establish novelty. This focus reflects the project's position as a research contribution grounded in two clearly defined methodological innovations, each with a well-bounded prior literature.

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